

Mian Usman

Lahore, Punjab, Pakistan

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Education

Lahore Garrison University

Oct 2022 – Jun 2026

Bachelor of Science in Software Engineering

Lahore, Pakistan

- Coursework: Machine Learning, Data Structures, Algorithms, Database Systems, Web Technologies, Big Data Analytics

KIPS Education System

Sep 2020 – Mar 2022

Intermediate in Computer Science (ICS)

Lahore, Pakistan

Professional Experience

Data Science & Web Development Intern

Aug 2025 – Sep 2025

Packages Group

Lahore, Pakistan

- Developed front-end interface for real-time AI detection system, integrating machine learning models with responsive web design
- Maintained Naqsh Art School's e-commerce platform, handling user authentication, product catalogue management, and POS/payment gateway integration.
- Collaborated with cross-functional teams to deliver AI-driven features and optimize user experience across web applications
- Implemented data visualization dashboards to track system performance and user engagement metrics

AI/ML Engineering Intern

Apr 2025 – Jun 2025

DevelopersHub Corporation

Lahore, Pakistan

- Designed and deployed end-to-end machine learning pipeline for fake news detection achieving 93% classification accuracy
- Implemented customer segmentation system using K-Means clustering, processing 10,000+ data points for targeted marketing insights
- Built sentiment analysis model using NLP techniques (tokenization, TF-IDF) and Logistic Regression for product review classification
- Developed Flask-based web interface and Tkinter desktop application for model deployment, ensuring seamless user interaction
- Resolved deployment challenges on Render cloud platform, optimizing model performance and reducing inference time by 30%
- Handled imbalanced datasets using SMOTE and data augmentation techniques to improve model robustness

Projects

Proactive Intrusion Detection System (PIDS) | Python, XGBoost, LightGBM,

Oct 2025 – June 2026

Django REST, React, PostgreSQL, Ollama llama3.1:2b, Scapy

- Architected four-stage detection pipeline: XGBoost binary classifier → LightGBM multi-class (14 attack types) → Connection Rate Detector → LLM Behaviour Engine (Ollama llama3.1:2b)
- Trained models on CICIDS dataset (15.7M rows, 6 GB) with GPU-accelerated XGBoost; engineered 31 statistical features from live packets using Scapy
- Built real-time WebSocket alert dashboard (React 18, Material-UI) with attack grouping, LLM re-analysis, and automated PDF threat reports
- Implemented JWT-based RBAC (5 roles, HMAC-SHA256), password policy enforcement, audit logging, and 40-issue security audit
- Documented system architecture, pluggable detection pipelines (XGBoost, LightGBM, Deep Neural Networks), and local LLM integration in an IEEE-format paper; published as *Predictive Intrusion Detection Using System Logs* on ResearchGate (DOI: 10.13140/RG.2.2.18965.69606)

CyberHive – Cyber Threat Intelligence Platform | *Hadoop, Spark MLlib, Hive, Apr 2026 – June 2026* *HBase, React*

- Designed big data platform processing ~15 GB of threat intelligence across 5 datasets (CSE-CIC-IDS2018, NVD/CVE, Abuse.ch, MaxMind GeoLite2, DNS logs)
- Implemented end-to-end pipeline using HDFS, MapReduce, YARN, Pig, Spark MLlib with a React threat visualisation dashboard

Satellite Image Classification | *Python, CNN, Transfer Learning Jun 2025*

- Developed CNN-based multi-class satellite image classifier using transfer learning with data augmentation and hyperparameter optimisation

Skills

Languages: Python, JavaScript (ES6+), SQL, C#

AI / ML: XGBoost, LightGBM, Scikit-learn, YOLOv8, ONNX, Pandas, NumPy, NLP (TF-IDF), CNN, K-Means, DBSCAN, SMOTE, LLM Integration (Ollama)

Web & APIs: Django REST Framework, React 18, Material-UI, Flask, Bootstrap, HTML5, CSS3, WebSockets, JWT, RBAC, RESTful APIs

Databases: PostgreSQL, MySQL, HBase, Hive

Big Data: Hadoop (HDFS, MapReduce, YARN), Spark MLlib, Flume, Sqoop, Pig

Security: Network Intrusion Detection, Scapy, Packet Analysis, Security Auditing

Tools: Git, GitHub, VS Code, Jupyter Notebook, Linux, Unity (C#)

Achievements & Certifications

Best Award in AI & Machine Learning Internship – DevelopersHub Corporation *Jun 2025*

Machine Learning, AI & Data Science Certification – Ehunar *Feb 2025*